

Patrick Thangarajah

SIMULATION FEA | TESTING VALIDATION | MECHANICAL DESIGN | PROTOTYPING

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Summary

Recent MASc graduate in mechanical engineering with practical experience designing, building, and testing hardware systems, including sensor-integrated test setups and control circuitry. Brings simulation, DOE, and data analysis skills to support designs from concept through validation.

Skills

Design & Prototyping	SolidWorks, Onshape, 3D Printing, Lathe, Mill, Bandsaw, Drill Press
Controls & Instrumentation	Arduino, Sensors, Motors, Data Acquisition, Circuit Design, Altium, LTspice
Testing & Analysis	DOE, JMP, Minitab, MATLAB, Microsoft Excel, Troubleshooting
Simulation & Programming	FEA, LS-DYNA, LS-PrePost, Python, C/C++
Education	

Master of Applied Science in Mechanical and Mechatronics Engineering

Waterloo, ON, Canada

University of Waterloo

Sep 2023 - Jul 2026

- Research in occupant safety and biomechanics

Bachelor of Applied Science in Mechanical Engineering, Co-op Program

Vancouver, BC, Canada

University of British Columbia

Sep 2018 - May 2023

- Biomechanics and Medical Devices Option

Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group

Waterloo, ON, Canada

University of Waterloo

Sep 2023 - Jul 2026

- Ran a 104-configuration LS-DYNA DOE and used JMP to quantify geometry effects on occupant response, identifying a configuration with 16% lower peak chest deflection
- Validated simulations against physical tests across two configurations and 10 response channels, achieving a 0.71 average ISO 18571 correlation score
- Troubleshoot model-to-test discrepancies in positioning, restraint fit, and response timing to improve agreement with physical tests
- Automated seat belt fitting with LS-PrePost scripts, reducing repeated setup steps by 43% (7 to 4)

Teaching Assistant - MTE 120: Circuits

Waterloo, ON, Canada

University of Waterloo

May 2024 - Aug 2024

- Supported laboratory instruction for a 100+ student circuits course, troubleshooting hardware and providing feedback on lab work and exams

R&D Engineer (Part-time)

Remote

Echoscope

May 2023 - Sep 2023

- Designed and iterated the bladder testing platform PCB, integrating pressure sensing, Arduino connections, power distribution, and motor and valve interfaces; refined component placement and routing and completed design-rule checks
- Authored five manufacturing and verification procedures covering PCB assembly, wiring, software setup, and functionality testing; all five were approved and used by the client

Summer Research Student

Vancouver, BC, Canada (Remote)

Centre for Heart Lung Innovation

May 2022 - Aug 2022

- Built a Jupyter Notebook transfer-learning script to diagnose diabetic kidney disease using an emerging imaging platform
- Improved diagnostic accuracy by 10% through model optimization, training refinement, and comparison with existing tissue-slide images

Mechanical Engineering Intern

Airdrie, AB, Canada

GN Corporation

Jan 2022 - Apr 2022

- Developed standardized work instructions and SOPs by reviewing engineering drawings, customer specifications, and existing manufacturing processes
- Reduced fiberglass cutting waste by 50% through process analysis
- Proposed a material-handling process improvement projected to reduce pick-and-place time by 50%

Engineering Intern

Victoria, BC, Canada

Breathe Global

Jul 2021 - Aug 2021

- Redesigned the analog control circuit for a negative-pressure breathing device in Altium, using LTspice simulation to select low-cost components
- Expanded the device's achievable breathing range from 70% to 88% of target

Projects

Effect of Seat Foam Properties on Injury Response in Rear Impact (ME 725 Course Project)

Waterloo, ON, Canada

University of Waterloo

Jan 2024 - Apr 2024

- Verified 5 polyurethane foam material models in LS-DYNA against input stress-strain response using single-element simulations, then implemented them in a rear-impact sled model with a Hybrid III ATD to calculate HIC15 and Nij injury metrics
- Determined that all 5 foams met the regulatory head injury limit but exceeded the regulatory neck injury limit at high speed, and that the softest foam reduced head injury risk by roughly 18–20% relative to the stiffest

Simulation of Cervical Fractures in American Football (ME 720 Course Project)

Waterloo, ON, Canada

University of Waterloo

Sep 2023 - Dec 2023

- Simulated head-first helmet impacts across three velocities (3.4–7.0 m/s) and three neck angles at the lowest speed condition using Hybrid III and GHBM human body models in LS-DYNA, then evaluated fracture force thresholds and injury location against peer-reviewed literature
- Discovered that predicted peak neck forces (2.26–2.34 kN) fell within the range reported across two cadaveric studies, and that the Neck Injury Criterion exceeded its injury-risk threshold in only 1 of 5 configurations despite fracture occurring in all 5

Soft Distensible Bladder Model - Capstone Project

Vancouver, BC, Canada

University of British Columbia

Sep 2022 - Apr 2023

- Prototyped bladder phantoms using silicone and agar, comparing pressure-volume response with literature to guide material selection
- Designed and built the control and data-acquisition interface for a pressurized soft-bladder test system, enabling real-time pressure/volume monitoring and automated filling/voiding cycles
- Validated system performance using integrated pressure sensing and controls, demonstrating stable and repeatable operation up to 4.75 kPa internal pressure

Payload Team Member

Vancouver, BC, Canada

UBC Unmanned Aircraft Systems (Student Design Team)

Sep 2019 - May 2021

- Designed and prototyped a payload drop system and rover in Onshape and SolidWorks, iterating through 3D printed proof-of-concept builds
- Reduced payload system weight by 50% by repurposing components from previous designs during concept development

Volunteer Researcher - Vacuum Injection Molding End-Effector (VIMEE) Gripper

Vancouver, BC, Canada

University of British Columbia Collaborative Advanced Robotics and Intelligent Systems(CARIS) Lab

Jan 2022 - Apr 2023

- Programmed Arduino-controlled LED signal patterns for pleat loading height and force feedback and contributed to user testing of the complete system

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- **Thangarajah, P.**, Watson, B., Tang, K., Tylko, S., & Cronin, D. (2025). *Analysis of rear seat shoulder belt anchor position on kinematics of small-stature female occupants in frontal impact*. Injury Biomechanics Symposium, The Ohio State University, Columbus, OH, USA, May 18–21, 2025.